Contest Problem Set 12221

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Name _____

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1. $122 + 22 =$ _____.
2. $479 + 342 =$ _____.
3. The hundreds digit of 58934 is _____.
4. $5 + 8 \times 5 =$ _____.
5. $712 - 389 =$ _____.
6. The remainder of $87 \div 5$ is _____.
7. $17 + 41 + 39 + 13 =$ _____.
8. $72 \div 12 =$ _____.
9. The product of 14 and 3 is _____.
10. (estimate) $258 + 473 + 559 =$ _____.
11. $42 \times 11 =$ _____.
12. $517 - 289 - 122 =$ _____.
13. $13^2 =$ _____.
14. The remainder of $107 \div 9$ is _____.
15. $15 \times 26 =$ _____.
16. $6 \times 9 \times 5 =$ _____.
17. $21 \times 19 =$ _____.
18. *CXXXIV* in Arabic numerals is _____.
19. $1.4 \times 0.9 =$ _____ (decimal).
20. (estimate) $199 \times 203 =$ _____.
21. The GCD of 12 and 20 is _____.
22. $38 \times 25 =$ _____.
23. $15 \times 8 \div 10 =$ _____.
24. If 1 mile is equal to 5280 feet, then $\frac{1}{10}$ mile is equal to _____ feet.
25. $273 \div 13 =$ _____.
26. The greater of $\frac{2}{3}$ and $\frac{5}{7}$ is _____ (fraction).
27. $27^2 =$ _____.
28. $\frac{2}{7} + \frac{1}{21} =$ _____ (fraction).
29. $14 + 17 + 20 + 23 + 26 =$ _____.
30. (estimate) $234567 \div 117 =$ _____.
31. $99 \times 13 =$ _____.
32. $\frac{1}{4} =$ _____ %.
33. The area of a square with perimeter 32 is _____.
34. $35^2 =$ _____.
35. The number of even whole numbers between 11 and 41 is _____.
36. $27 \times 23 =$ _____.
37. The LCM of 12 and 20 is _____.
38. 20% of 60 is _____.
39. 42 days is _____ weeks.
40. (estimate) $667 \times 180 =$ _____.

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41. $14 \times 4 + 8 \times 13 =$ _____.

42. The remainder of $474 \div 11$ is _____.

43. $87 \times 27 =$ _____.

44. The area of a right triangle with legs of length 12 and 9 is _____.

45. $54 \times 101 =$ _____.

46. The tenth term in the arithmetic sequence $3, 7, 11, \ldots$ is _____.

47. $22^2 - 12^2 =$ _____.

48. 45_6 in base 10 is _____.

49. $\sqrt{1156} =$ _____.

50. (estimate) $12 \times 142857 =$ _____.

51. $104 \times 107 =$ _____.

52. $12\frac{1}{2}\% =$ _____ (fraction).

53. 21_4 in base 2 is _____ $_2$.

54. $2^7 =$ _____.

55. The sum of the terms of the arithmetic sequence $5, 10, 15, \ldots, 100$ is _____.

56. $98 \times 95 =$ _____.

57. The measure of an interior angle in a regular quadrilateral is _____ $^\circ$.

58. $31 \times 111 =$ _____.

59. The median of the list $5, 11, 4, 13, 1$ is _____.

60. (estimate) $\sqrt{202222} =$ _____.

61. $4\frac{1}{3} \times 4\frac{2}{3} =$ _____ (mixed number).

62. Two fair coins are flipped. The probability that both show the same result is _____ (fraction).

63. $0.\overline{18} =$ _____ (fraction).

64. The number 56 written in base 5 is _____ $_5$.

65. $74 \times 125 =$ _____.

66. $63^2 =$ _____.

67. The perimeter of a right triangle with legs of length 8 and 15 is _____.

68. $15 \times 35 \times 21 =$ _____.

69. If $w = 7$, then $w^2 + 6w + 9 =$ _____.

70. (estimate) $51^3 =$ _____.

71. $988 \times 989 =$ _____.

72. The number of subsets of $\{L, E, T, S, G, O\}$ is _____.

73. $\sqrt[3]{636056} =$ _____.

74. The number of positive whole number divisors of 28 is _____.

75. If $3^x = \frac{1}{3}$, then $3^{2-x} =$ _____.

76. $\sqrt{8} \times \sqrt{98} =$ _____.

77. The sum of the terms of the infinite geometric sequence $1, \frac{1}{2}, \frac{1}{4}, \ldots$ is _____.

78. $75 \times 92 \times 9 =$ _____.

79. The surface area of a cube with an edge of length 8 is _____.

80. (estimate) $\sqrt{21 \times 22 \times 23} =$ _____.

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1.	A	B	C	D	E	11.	A	B	C	D	E	21.	A	B	C	D	E
2.	A	B	C	D	E	12.	A	B	C	D	E	22.	A	B	C	D	E
3.	A	B	C	D	E	13.	A	B	C	D	E	23.	A	B	C	D	E
4.	A	B	C	D	E	14.	A	B	C	D	E	24.	A	B	C	D	E
5.	A	B	C	D	E	15.	A	B	C	D	E	25.	A	B	C	D	E
6.	A	B	C	D	E	16.	A	B	C	D	E	26.	A	B	C	D	E
7.	A	B	C	D	E	17.	A	B	C	D	E	27.	A	B	C	D	E
8.	A	B	C	D	E	18.	A	B	C	D	E	28.	A	B	C	D	E
9.	A	B	C	D	E	19.	A	B	C	D	E	29.	A	B	C	D	E
10.	A	B	C	D	E	20.	A	B	C	D	E	30.	A	B	C	D	E

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1. Which of the following expressions has the greatest value?

(A) 12×5 (B) 2×29 (C) 4×14 (D) 6×9 (E) 3×19

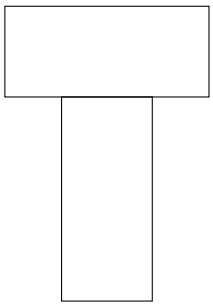
2. At the theater concession stand, a large bucket of popcorn is sold for \$5, and a regular bucket of popcorn is sold for \$3. A group celebrating a birthday purchased three large buckets of popcorn and two regular buckets of popcorn. How many dollars did the group spend on popcorn?

(A) \$22 (B) \$21 (C) \$20 (D) \$19 (E) \$18

3. Mackenzie started watching the most recent episode of *The Bandamorian* at 7:48 PM. The episode was 44 minutes long, and Mackenzie watched it from start to finish without taking any breaks. What time was it when Mackenzie finished watching the episode?

(A) 8:32 PM (B) 8:28 PM (C) 8:22 PM (D) 8:24 PM (E) 8:34 PM

4. The *T*-shaped figure below is formed by two identical rectangles, each with a width of 9 and a length of 4. What is the perimeter of the *T*-shaped figure?



(A) 44 (B) 48 (C) 42 (D) 40 (E) 52

5. If the day before yesterday was a Tuesday, then what day of the week will it be two days after tomorrow?

(A) Friday (B) Thursday (C) Saturday (D) Monday (E) Sunday

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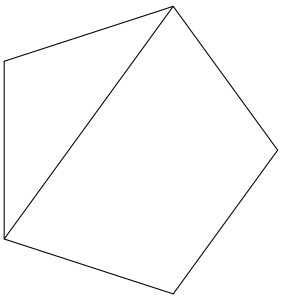
6. Britney handed the cashier at the arcade three $20 bills and asked to be paid back the same amount using only bills worth less than $20. The cashier handed Britney three $10 bills, four $5 bills, and two rare $2 bills. The remaining bills were all $1 bills. How many $1 bills did Britney receive from the cashier?

(A) 6 (B) 1 (C) 11 (D) 4 (E) 9

7. How many positive whole numbers greater than 1 and less than 10 evenly divide 168?

(A) 7 (B) 6 (C) 3 (D) 5 (E) 4

8. A regular pentagon is shown below, with one diagonal drawn. When all of the possible diagonals of the pentagon are drawn, at how many different points inside the pentagon do two or more diagonals intersect, or cross another diagonal?



(A) 8 (B) 5 (C) 6 (D) 4 (E) 10

9. Five friends enjoyed a dinner together at a restaurant, and planned to split the bill evenly. Unfortunately, Thomas forgot his money. The other four guests paid an extra $6 each to cover Thomas' share of the bill. What was the total bill for the meal for all five friends?

(A) $140 (B) $150 (C) $120 (D) $100 (E) $90

10. The first mathleague.org Elementary National Championship was held in 2014, and the championship has been held every year since. Jake turned eleven years old the same year he competed in the eighth mathleague.org Elementary National Championship. What year was Jake born?

(A) 2013 (B) 2012 (C) 2010 (D) 2011 (E) 2009

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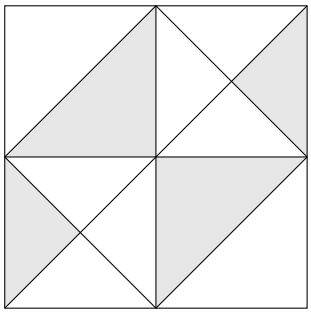
11. The sum of three different prime numbers is 32. What is the greatest possible value for the product of the three numbers?

(A) 504 (B) 546 (C) 322 (D) 442 (E) 418

12. The figure below shows a square, with one diagonal of the square drawn. A second square is formed by connected the midpoints of the sides of the original square, with both diagonals of that smaller square drawn as well. If the area of the original square is 24, then what is the total area of the shaded regions?



(A) 12 (B) 6 (C) 8 (D) 10 (E) 9

13. Four soccer players are choosing an order to try penalty kicks, but Alejandro refuses to be first or last, and Scott refuses to be first. How many ways can the four soccer players be placed in an order to try penalty kicks?

(A) 12 (B) 6 (C) 4 (D) 8 (E) 9

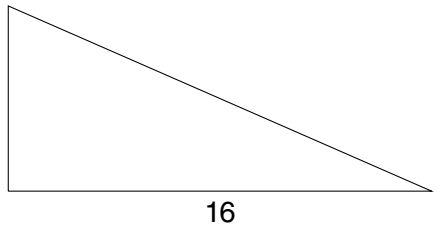
14. A number is twice as far from 51 on the number line as it is from 75. What is the sum of all possible values of the number?

(A) 99 (B) 67 (C) 166 (D) 190 (E) 142

15. Raymond averaged a score of 27 on his four quizzes. Raymond's scores on the first two quizzes were equal, and his score on the third quiz was six points less than his score on the fourth quiz. Raymond scored a 29 on his fourth quiz. What was Raymond's score on each of the first two quizzes?

(A) 25 (B) 28 (C) 29 (D) 27 (E) 26

16. The right triangle below has one leg of length 16 and an area of 56. What is the least positive whole number that is greater than the length of the hypotenuse of the triangle?



(A) 20 (B) 17 (C) 19 (D) 18 (E) 21

17. Shana chose a positive whole number and divided it 65. Her result was greater than $\frac{1}{8}$ but less than $\frac{1}{3}$. How many different positive whole numbers could Shana have chosen?

(A) 11 (B) 13 (C) 14 (D) 10 (E) 12

18. The factorial of a positive whole number is marked by a ! after the number, and the value of the factorial of a positive whole number is the product of all of the positive whole numbers less than or equal to the number. For example, $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$, which is 120. What is the value of $\frac{11!-10!-9!}{8!}$?

(A) 861 (B) 891 (C) 900 (D) 810 (E) 899

19. The Little Six basketball league has six teams. Each team in the league plays four games against teams from other leagues. Then each team in the league plays every other team in the league twice. How many total games are played where at least one of the teams playing is from The Little Six basketball league?

(A) 64 (B) 72 (C) 84 (D) 60 (E) 54

20. If $A \heartsuit B = \frac{A+2B}{2}$, then for what number k is $10 \heartsuit (6 \heartsuit k)$ equal to 19?

(A) 11 (B) 13 (C) 14 (D) 12 (E) 15

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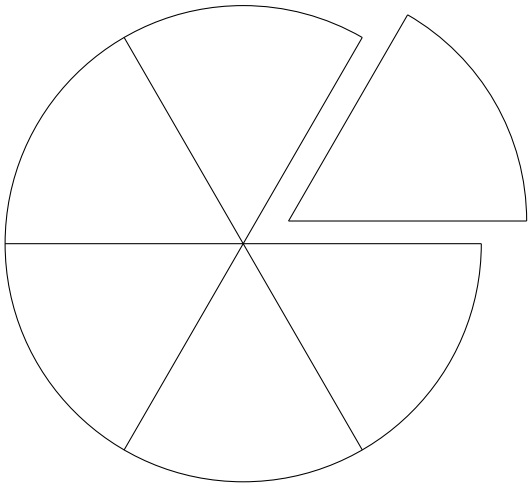
21. A bag contains only red, blue, and green marbles. The ratio of red marbles to blue marbles is $15 : 16$. The ratio of green marbles to blue marbles is $5 : 4$. After one dozen blue marbles are added to the bag, the ratio of red marbles to blue marbles is $5 : 6$, and the ratio of green marbles to blue marbles is $10 : 9$. What is the total number of red and green marbles in the bag?

(A) 190 (B) 180 (C) 210 (D) 195 (E) 200

22. Which of the following is greater than 20?

(A) $9\sqrt{5}$ (B) $8\sqrt{6}$ (C) $6\sqrt{11}$ (D) $4\sqrt{23}$ (E) $14\sqrt{2}$

23. At Pete's Pastel Pizzeria, an extra large pizza is 21 inches in diameter. The pizza is sliced into six equal pieces by making three cuts along diameters of the pizza, as shown below. To the nearest whole number of inches, what is the perimeter of one of these six pieces of pizza?



(A) 35 (B) 31 (C) 33 (D) 36 (E) 32

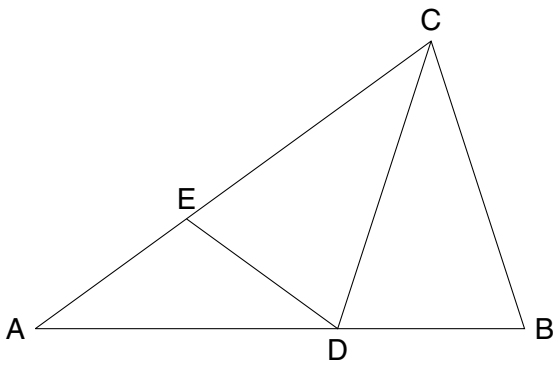
24. The value of the decimal product $2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3 \cdot 7$ is written in base 5. The result has how many more digits that are 4 than digits that are 0?

(A) 0 (B) 3 (C) 5 (D) 2 (E) 1

25. On Monday, Travis averaged 45 miles per hour on his roundtrip journey from home to work and then back home. On Tuesday, Travis averaged 40 miles per hour on the same roundtrip journey, and his roundtip journey took 12 minutes longer than it did on Monday. How far, in miles, is Travis' home from his work?

(A) 36 (B) 72 (C) 45 (D) 54 (E) 48

26. In the figure below, segments *AE*, *ED*, and *DB* have equal length, and segments *CE*, *CD*, and *CB* have equal length. Additionally, points *A*, *E*, and *C* are collinear, and points *A*, *D*, and *B* are collinear. What is the sum of the digits of the angle measure, in degrees, of angle *EAD*?



(A) 12 (B) 3 (C) 8 (D) 6 (E) 9

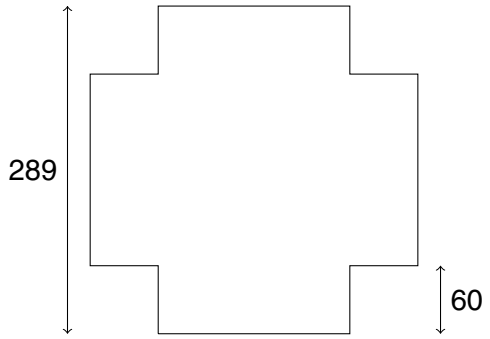
27. Rudy enters the number 1892 into his calculator and presses the square root button. He then adds 1892 to the result, and presses the square root button again. He repeats the process a dozen more times, always adding 1892 to the result of the square root and then pressing the square root button. The final number displayed on Rudy's calculator is nearest to what positive whole number?

(A) 47 (B) 46 (C) 45 (D) 44 (E) 43

28. What is the sum of the digits of the greatest four-digit positive whole number that has a remainder of 4 when divided by 7, a remainder of 5 when divided by 9, and a remainder of 6 when divided by 11?

(A) 24 (B) 20 (C) 21 (D) 23 (E) 25

29. A dodecagon is formed by removing four squares of side length 60 from a square of side length 289, as shown below. What is the side length of the largest possible square that lies entirely on or in this dodecagon?



(A) 224 (B) 216 (C) 221 (D) 220 (E) 225

30. Three points are chosen at random on a circle. What is the probability that the triangle formed by the three points is acute?

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{8}$ (E) $\frac{1}{4}$

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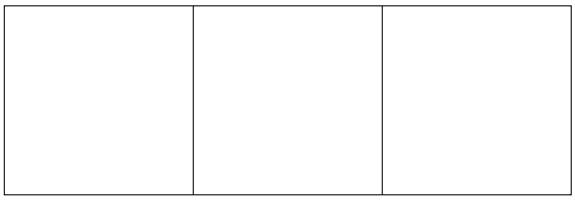
Problems 1 & 2

1. On Monday, Kenneth's walk to school was twelve minutes. On Tuesday, his walk to school was eleven minutes. On Wednesday, his walk to school was thirteen minutes. On Thursday, his walk to school was one minute longer than his walk on Wednesday, and on Friday his walk to school was two minutes shorter than his walk on Monday. Altogether, from Monday through Friday, how many minutes did Kenneth spend walking to school?

1.

2. A rectangular piece of paper on a table is divided into three identical squares, as shown below. Each of the squares is then colored either red or blue. If rotations of the same coloring are considered to be identical, then how many different ways are there to color the three squares?

2.



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Problems 3 & 4

3. Quarters are coins worth 25 cents, and dimes are coins worth 10 cents. Vai has a collection of seven quarters and six dimes. What is the fewest number of additional coins, all quarters or dimes, that Vai needs so that her quarters and dimes are worth a total of exactly 400 cents?

3.

4. Jamie runs 18 meters in the same time Jessica runs 15 meters. Jamie and Jessica are standing next to each other on a circular running track that is 360 meters in length, and both start running at the same time and in the same direction. If Jamie and Jessica each run at an unchanging pace, how many laps will Jamie have completed around the track when she passes Jessica for the first time?

4.

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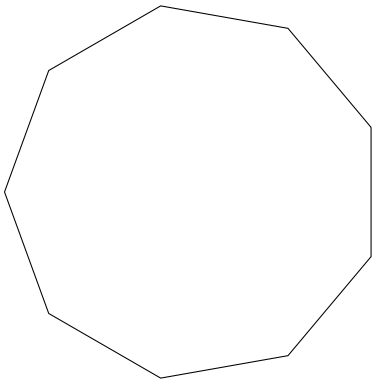
Problems 5 & 6

5. A number is called a *perfect square* if it is the result of a whole number multiplied by itself. For example, since $5 \times 5 = 25$, the number 25 is a perfect square. What is the four-digit perfect square that has a thousands digit less than its hundreds digit, a hundreds digit less than its tens digit, and a tens digit less than its ones digit?

5.

6. A regular nonagon is shown below. How many different triangles have vertices that are also vertices of the nonagon, and share at least one side with the nonagon?

6.



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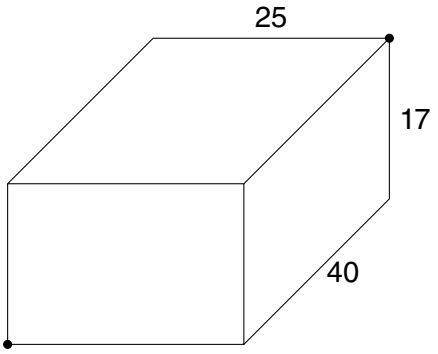
Problems 7 & 8

7. Emily needed to read a book over the weekend. On Friday after school, she started reading the book from the beginning. That evening she read half the pages of the book plus an additional twelve pages. On Saturday morning, she read half the remaining pages plus an additional nine pages. On Sunday, she read the final forty-seven pages of the book. How many pages long was the book?

7.

8. An ant travels along the surface of a $17 \times 25 \times 40$ right rectangular prism, starting at one corner of the prism and ending at the corner that is farthest away from the starting corner. What is the shortest possible distance the ant traveled? Express your answer to the nearest whole number.

8.



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Team Round 12221

School or Team

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Name _____

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1.	2.	3.	4.	5.
6.	7.	8.	9.	10.

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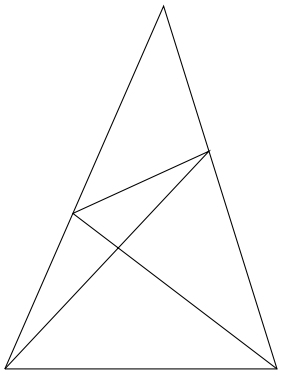
1. What is the sum of all of the positive whole numbers less than 31 that are even but not multiples of 3?

2. Marlee has a collection of memes saved on her smartphone. If Marlee had one more meme, the memes could be displayed in a gallery with seven memes per row and no incomplete rows. If Marlee had four fewer memes, the memes could be displayed in a gallery with six memes per row and no incomplete rows. What is the least possible number of memes that Marlee could have on her smartphone?

3. A twelve-hour analog clock rings its chimes every hour, and the chimes ring a number of times equal to the number of the hour. For example, at both seven o'clock in the morning and seven o'clock in the evening, the chimes ring seven times. This is true for all hours except midnight. At midnight, to signal the start of a new day, the chimes ring two dozen times. How many total times do the chimes ring in the month of October?

4. Using only segments in the figure below as sides of the triangle, how many different triangles can be formed?



5. Each of the digits 2, 4, 7, and 9 is used exactly once to replace a $\square$ in the expression $(\square + \square) \times (\square + \square)$. For example, one possible value of the expression is $(2 + 7) \times (4 + 9) = 108$. What is the value of the greatest possible value of the expression minus the least possible value of the expression?

6. A jar of salad dressing is filled with a mix of 80% oil and the rest is vinegar. A second jar of salad dressing, half the size of the first jar, is filled with a mix of 90% oil and the rest is vinegar. The contents of both jars are mixed in a larger jar. To the nearest whole number, what percent of the resulting mixture is vinegar?

7. A palindrome is a number that is unchanged when the order of its digits is reversed. For example, 1221 and 6556 are both palindromes. How many four-digit positive whole numbers are palindromes and multiples of 99?

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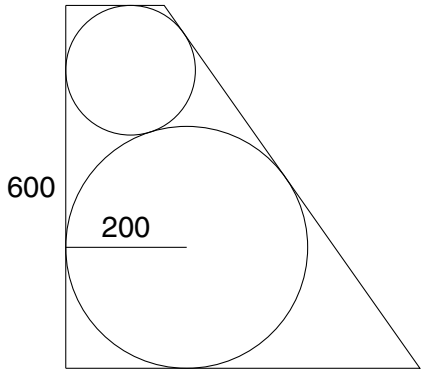
8. Six students are going to be seated around a circular table. Andre insists on sitting next to Bryce, but Bryce refuses to sit next to Charlotte. If seating arrangements are considered different only if a student has someone different seated beside them on their left, then how many different ways can the students be seated around the table?

9. In a trapezoid, a leg of length 600 is perpendicular to both bases. Two circles are each tangent to three sides of the trapezoid as shown. The radius of the larger circle is 200. To the nearest whole number, what is the radius of the smaller circle?



10. Leland has a large collection of identical toy bricks that are each 1 unit tall, and a large collection of identical toy bricks that are each 2 units tall. Using these bricks, how many different ways can Leland assemble a tower that is fifteen units tall?

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Answers 12221

Sprint Round

1. A
2. B
3. A
4. A
5. E
6. A
7. B
8. B
9. C
10. C
11. D
12. E
13. D
14. C
15. B
16. D
17. B
18. B
19. E
20. A
21. C
22. A
23. E
24. B
25. A
26. E
27. D
28. D
29. C
30. E

Target Round

1. 60
2. 6
3. 9
4. 6
5. 1369
6. 54
7. 248
8. 58

Team Round

1. 150
2. 34
3. 5208
4. 12
5. 25
6. 17(%)
7. 10
8. 36
9. 107
10. 987

Number Sense

1. 144
2. 821
3. 9
4. 45
5. 323
6. 2
7. 110
8. 6
9. 42
10. $[1226, 1354]$
11. 462
12. 106
13. 169
14. 8
15. 390
16. 270
17. 399
18. 134
19. 1.26
20. $[38378, 42416]$
21. 4
22. 950
23. 12
24. 528
25. 21
26. $\frac{5}{7}$
27. 729
28. $\frac{1}{3}$
29. 100
30. $[1905, 2105]$
31. 1287
32. 25
33. 64
34. 1225
35. 15
36. 621
37. 60
38. 12
39. 6
40. $[114057, 126063]$
41. 160
42. 1
43. 2349
44. 54
45. 5454
46. 39
47. 340
48. 29
49. 34
50. $[1628570, 1799998]$
51. 11128
52. $\frac{1}{8}$
53. 1001
54. 128
55. 1050
56. 9310
57. 90
58. 3441
59. 5
60. $[428, 472]$
61. $20\frac{2}{9}$
62. $\frac{1}{2}$
63. $\frac{2}{11}$
64. 211
65. 9250
66. 3969
67. 40
68. 11025
69. 100
70. $[126019, 139283]$
71. 977132
72. 64
73. 86
74. 6
75. 27
76. 28
77. 2
78. 62100
79. 384
80. $[98, 108]$

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Solutions 12221

Sprint Round Solutions

1. Evaluating each, $6 \times 9 = 54$, $4 \times 14 = 56$, $3 \times 19 = 57$, $2 \times 29 = 58$, and $12 \times 5 = 60$. The greatest value among these is $\boxed{60}$.

2. The large buckets of popcorn totaled $3 \times \$5 = \$15$, and the regular buckets of popcorn totaled $2 \times \$3 = \$6$. The total price of the popcorn was $\$15 + \$6 = \boxed{\$21}$.

3. After 12 minutes have passed, it will be 8:00 PM, and there will be $44 - 12 = 32$ minutes left of the episode. After those 32 minutes elapse, the time will be $\boxed{\text{8:32 PM}}$.

4. The perimeter of each individual rectangle is $4 + 4 + 9 + 9 = 26$. If the two rectangles were separate, their total perimeter would be $26 + 26 = 52$. But when the *T*-shaped figure is formed, the segments of length 4 where the rectangles meet are removed from the figure. The perimeter of the *T*-shaped figure is therefore $52 - 4 - 4 = \boxed{44}$.

5. If the day before yesterday was a Tuesday, then yesterday was a Wednesday, and today is a Thursday. Tomorrow is a Friday. One day after tomorrow is a Saturday, and two days after tomorrow is a $\boxed{\text{Sunday}}$.

6. The total amount of money Britney gave the cashier was $3 \times \$20 = \$60$. The amount of money Britney received from the cashier in bills other than \$1 bills was $3 \times \$10 + 4 \times \$5 + 2 \times \$2 = \$54$. Therefore, Britney received $60 - 54 = \boxed{6}$ \$1 bills.

7. The number 168 can be expressed as 21×8. Since it is divisible by 8, it is also divisible by 2 and 4. Since it is divisible by 21, it is divisible by 3 and 7. Since it is divisible by 2 and 3, it is also divisible by 6. It is not divisible by 5, and since $1 + 6 + 8 = 15$ is not a multiple of 9, it is not divisible by 9. The whole numbers greater than 1 and less than 10 that evenly divide 168 are 2, 3, 4, 6, 7, and 8, for a total of $\boxed{6}$ divisors.

8. When all of the 5 diagonals are drawn, a five-pointed star is formed, and each diagonal intersects exactly 2 other diagonals. This is $5 \times 2 = 10$ intersections, but it double counts each intersection, so the number of points of intersection is $10 \div 2 = \boxed{5}$.

9. The other four friends contributed a total of $\$6 \times 4 = \$24$ for Thomas, which was his portion of the bill. Therefore, the total bill for all five friends was $5 \times \$24 = \boxed{\$120}$.

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10. The first championship was held in 2014, so the eighth championship was held in $2014+(8-1) = 2021$. Since Jake turned 11 years old in 2021, he was born in $2021 - 11 = \boxed{2010}$.

11. The sum of three odd numbers is an odd number, but since 32 is even, at least one of the primes must be even. The only even prime is 2, so the other two primes are odd and sum to $32 - 2 = 30$. The possible prime pairs that sum to 30 are 7 and 23, 11 and 19, and 13 and 17. The products of each pair are 161, 209, and 221, so the greatest possible value for the product of the three numbers is $2 \cdot 221 = \boxed{442}$.

12. The two large triangular regions are each $\frac{1}{8}$ of the original square, and the two small triangular regions are each $\frac{1}{16}$ of the original square. Altogether, the shaded regions total $\frac{1}{8} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16} = \frac{3}{8}$ of the square, and have an area of $\frac{3}{8} \cdot 24 = \boxed{9}$.

13. Since neither Alejandro or Scott wants to be first, there are only 2 possibilities for the player that goes first. After selecting this player, because Alejandro does not want to be last, there are only 2 possibilities for the player that goes last. After the first and last player are selected, there are 2 possibilities for the player that goes second, as Alejandro is now a possibilitity, and then, after all of these players are selected, only 1 possibility for the player that goes third. The total number of possible orderings of the players is $2 \times 2 \times 2 \times 1 = \boxed{8}$.

14. There are two possibilities for the location of the number on the number line. The first possibility is that the number is between 51 and 75. The distance from 75 to 51, which is $75 - 51 = 24$, would need to be split into two pieces, one twice the other. Since $\frac{2}{3}$ of 24 is 16, the number could be $51 + 16 = 67$. The second possibility is that the number is greater than 75, and the distance from the number to 51 would always be 24 more than the distance from the number to 75, so for that distance to be double the distance from 75, the distance from 75 would also be 24, and the number could be 99. Therefore, the sum of all possible values of the number is $67 + 99 = \boxed{166}$.

15. The total points Raymond scored on the four quizzes was $27 \times 4 = 108$. Raymond scored a 29 on the fourth quiz and a $29 - 6 = 23$ on the third quiz. Therefore the sum of Raymond's scores on his first two quizzes was $108 - 29 - 23 = 56$, and since both scores were equal, each was $56 \div 2 = \boxed{28}$.

16. Because the area of the triangle is 56, the product of the lengths of the legs is $2 \cdot 56 = 112$, so the length of the other leg is $\frac{112}{16} = 7$. By the Pythagorean Theorem, the square of the length of the hypotenuse is $16^2 + 7^2 = 305$. Since $17^2 = 289$ and $18^2 = 324$, the least positive whole number that is greater than the length of the hypotenuse is $\boxed{18}$.

17. If the number divided by 65 was greater than $\frac{1}{8}$, then 8 times the number must be greater than 65. This is true for all numbers greater than or equal to 9. Similarly, if the number divided by 65 was less than $\frac{1}{3}$, then 3 times the number must be less than 65. This is true for all numbers less than or equal to 21. Therefore Shana could have chosen any positive whole number from 9 through 21, so the number of possibilities is $21 - 9 + 1 = \boxed{13}$.

18. The number 11! can be expressed as $11 \cdot 10 \cdot 9 \cdot 8!$. Similarly, the number 10! can be expressed as $10 \cdot 9 \cdot 8!$, and 9! as $9 \cdot 8!$. Then $\frac{11!-10!-9!}{8!} = \frac{11 \cdot 10 \cdot 9 \cdot 8! - 10 \cdot 9 \cdot 8! - 9 \cdot 8!}{8!}$, which is $11 \cdot 10 \cdot 9 - 10 \cdot 9 - 9$. Factoring yields $9 \cdot (11 \cdot 10 - 10 - 1)$, or $9 \cdot 99$, which is $\boxed{891}$.

19. The number of games played against teams in other leagues is $6 \times 4 = 24$. Now suppose that of the two games between any two teams in the league, each team is designated as the home team for exactly one of the games. Then each team will have exactly $6 - 1 = 5$ games as the home team. Therefore there are $6 \cdot 5 = 30$ games played between two teams in the league, and the total number of games played by teams in the Little Six is $24 + 30 = \boxed{54}$.

20. The expression $A \heartsuit B$ can be simplified to $\frac{A}{2} + B$. So if $10 \heartsuit m = 19$, then $5 + m = 19$, and $m = 14$. Then $6 \heartsuit k = 14$, so $3 + k = 14$, and $k = \boxed{11}$.

21. Let the number of blue marbles be $16b$. Then the number of red marbles is $15b$, and the number of green marbles is $20b$. After adding 12 blue marbles to the bag, the ratio of red marbles to blue marbles is $\frac{15b}{16b+12}$, which is $\frac{5}{6}$. Setting these quantities equal and simplifying, $90b = 80b + 60$, so $10b = 60$ and $b = 6$. Then the number of red marbles if $15 \cdot 6 = 90$, the number of green marbles is $20 \cdot 6 = 120$, and the total number of red and green marbles is $90 + 120 = \boxed{210}$.

22. The square of 20 is $20^2 = 400$. Squaring each answer choice yields $4^2 \cdot 23 = 368$, $8^2 \cdot 6 = 384$, $14^2 \cdot 2 = 392$, $6^2 \cdot 11 = 396$, and $9^2 \cdot 5 = 405$. The only answer choice that is greater than 400 when squared, and therefore greater than 20, is $\boxed{9\sqrt{5}}$.

23. The perimeter of a pizza slice is composed of two straight edges, each with a length equal to the radius of the pizza, and a curved portion that is equal to $\frac{1}{6}$ of the circumference of the pizza. The two radii sum to the diameter of the pizza, which is 21. Using $\frac{22}{7}$ as an approximation for π, the curved portion has length of very nearly $\frac{1}{6} \cdot \frac{22}{7} \cdot 21 = 11$. To the nearest whole number of inches, the total perimeter is $21 + 11 = \boxed{32}$.

24. The product can be grouped as $(2 \cdot 2 \cdot 2 \cdot 3) \cdot (2 \cdot 3 \cdot 3 \cdot 7)$, or $24 \cdot 126$. Converting each to base 5 yields $44_5 \cdot 1001_5$. This is $44_5 \cdot 1000_5 + 44_5 \cdot 1$, or $44000_5 + 44_5 = 44044_5$. The number of digits that are 4 exceeds the number of digits that are 0 by $\boxed{3}$.

25. Let D be the distance, in miles, from Travis' home to his work. Then his roundtrip journey is distance $2D$. Let T be the time, in hours, of Travis' roundtrip on Monday. Then $T + \frac{12}{60}$, or $T + \frac{1}{5}$, is the time, in hours, of Travis' roundtrip on Tuesday. Since rate is equal to distance divided by time, $\frac{2D}{T} = 45$, and $\frac{2D}{T+\frac{1}{5}} = 40$. Simplifying both equations, $2D = 45T$, and $2D = 40T + 8$. Therefore, $45T = 40T + 8$, and $T = \frac{8}{5}$. Then $2D = \frac{8}{5} \cdot 45$, so $2D = 72$, and D is $\boxed{36}$.

26. Triangles ECD and DCB are isosceles and congruent, and triangle AED is isosceles. Let the degree measure of angle CDB be x. Then the degree measure of angle EDC is also x, and the degree measure of angle ADE is $180^\circ - 2x$. But angle CED is also x, so angle AED is $180^\circ - x$, and because triangle AED is isosceles, the degree measure of angle ADE is $\frac{180^\circ-(180^\circ-x)}{2}$. Setting these two values for the measure of angle ADE equal, $180^\circ - 2x = \frac{180^\circ-(180^\circ-x)}{2}$, so $360^\circ - 4x = x$, and $x = 72^\circ$. Then the measure of angle EAD, which is equal to the measure of angle ADE, is $180^\circ - 2 \cdot 72^\circ = 36^\circ$, and $3 + 6 = \boxed{9}$.

27. Suppose there is a number k to which this operation converges. That is, a number that the results on Rudy's calculator approach more and more closely the more he performs the operation. Then $\sqrt{1892 + k} = k$. Squaring both sides of this equation, $1892 + k = k^2$, and $k^2 - k - 1892 = 0$. This quadratic can be factored as $(k - 44)(k + 43)$. Since the value of k is the result of a square root, k must be positive, so k is 44. A dozen applications of the operation are far more than enough to get very close to this value, so the positive whole number nearest to Rudy's result is $\boxed{44}$.

28. Considering the first two conditions, the smallest number that has a remainder of 4 when divided by 7 and a remainder of 5 when divided by 9 is 32. Since the least common multiple of 7 and 9 is 63, any positive whole number that meets these two conditions is of the form $32 + 63k$, where k is a non-negative whole number. For $32 + 63k$ to be 6 more than a multiple of 11, $26 + 63k$ must be a multiple of 11, and since $22 + 55k$ is always a multiple of 11, $4 + 8k$ must be a multiple of 11. This first occurs when k is 5. Therefore the smallest number that meets all three conditions is $5 \cdot 63 + 32 = 347$. Since $11 \cdot 63 = 693$ is the least common multiple of 7, 9, and 11, any number that meets all three conditions is of the form $347 + 693k$. The greatest possible value of k for which this is a four-digit number is $k = 13$, yielding $347 + 693 \cdot 13 = 9356$, and $9 + 3 + 5 + 6 = \boxed{23}$.

29. The longer side length of the dodecagon is $289 - 2 \cdot 60 = 169$. The largest possible square has one vertex on each of the four sides of length 169, and sides of the square that just touch the vertices of the dodecagon where two side lengths of 60 meet. Now partition the dodecagon into a central square of side length 169 and four congruent rectangles, each on a side of this central square, that are 60×169. The sides of the largest possible square partition each of these 60×169 rectangles into three right triangles. The largest of these triangles is half of the rectangle, and has a hypotenuse of length 169 and an altitude to the hypotenuse of length 60. The side of the square is equal to the sum of the lengths of the legs of this right triangle. Let the lengths of these two legs be x and y. By the Pythagorean Theorem, $x^2 + y^2 = 169^2$. By the area of the triangle, $xy = 60 \cdot 169$. Then $x^2 + 2xy + y^2 = 169^2 + 2 \cdot 60 \cdot 169$, or $289 \cdot 169$. Therefore, $x + y$, which, again, is the side length of the square, is $\sqrt{289 \cdot 169} = 17 \cdot 13$, or $\boxed{221}$.

30. The triangle is acute only if all three points do not lie on the same semicircle. Any two points will lie on a semicircle with probability 1. If the two points are the endpoints of a minor arc of 180°, then the third point will lie on a semicircle with the other two with probability 1. If the two points are the endpoints of a minor arc of k°, any point in the k° minor arc will lie in the same semicircle, as will any point in the $180^\circ - k^\circ$ arc on either end of the minor arc. This probability is $\frac{k^\circ+180^\circ-k^\circ+180^\circ-k^\circ}{360^\circ} = \frac{360^\circ-k^\circ}{360^\circ}$. Since k° varies from 0° to 180°, the probability all three points lie on a semicircle varies linearly from 1 to $\frac{1}{2}$, and the average probability is $\frac{1+\frac{1}{2}}{2} = \frac{3}{4}$. This is the probability the three points lie on a semicircle and form an obtuse triangle, so the probability the three points do not lie on a semicircle and form an acute triangle is $1 - \frac{3}{4} = \boxed{\frac{1}{4}}$.

Target Round Solutions

1. The Monday walk was 12 minutes, Tuesday was 11, and Wednesday was 13. Thursday was $13 + 1 = 14$ minutes, and Friday was $12 - 2 = 10$ minutes. Altogether, the total number of minutes was $12 + 11 + 13 + 14 + 10 = \boxed{60}$.

2. The squares could be colored all red, or all blue, for 2 colorings. The only other possibility is for two of the squares to be colored red or two to be colored blue. If two are colored red, then the blue square can either be in the middle or on the end, for 2 colorings. Similarly, if two are colored blue, then the red square can either be in the middle or on the end, for another 2 colorings. Altogether, the total number of possible colorings is $2 + 2 + 2 = \boxed{6}$.

3. Vai's quarters are worth $7 \times 25 = 175$ cents, and her dimes are worth $6 \times 10 = 60$ cents. Vai has a total of $175 + 60 = 235$ cents, and needs $400 - 235 = 165$ cents more. The fewest number of coins is accomplished by maximizing the number of quarters used. While 6 quarters are worth 150 cents, the remaining $165 - 150 = 15$ cents cannot be formed using quarters and dimes. But 5 quarters are worth 125 cents, and the remaining $165 - 125 = 40$ cents can be formed with 4 dimes. The total number of additional coins needed is $5 + 4 = \boxed{9}$.

4. Regardless of the distance run, Jamie will always run $\frac{18}{15} = \frac{6}{5}$ as far as Jessica can in the same amount of time. When Jessica runs 1 lap, Jamie will have run $\frac{6}{5}$ of a lap. When Jessica runs 5 laps, Jamie will pass her for the first time, and her number of laps run will be $\boxed{6}$.

5. The units digit of a perfect square can only be a 1, 4, 5, 6, or 9, and if the units digit of a perfect square is 5, then the tens digit must be 2. Based on the inequality conditions for digits, we find that the units digit can not equal 1 or 5 because there is no way to select the hundreds and thousands digits. Additionally, the only number that ends in 4 that satisfies the inequality condition for digits is 1234, but that is not a perfect square. If the units digit is a 6, then the number has to be less than 3000 because 3456 is not a perfect square, but $34^2 = 1156$, $36^2 = 1296$, $44^2 = 1936$, $46^2 = 2116$, $54^2 = 2916$, and $56^2 = 3136$ fail the digit condition, so the units digit cannot be a 6. If the units digit is a 9, the number must be less than 6000 because 6789 is not a perfect square. Checking squares with a units digit of 9, $33^2 = 1089$, and $37^2 = 1369$. All of 43^2, 47^2, 53^2, 57^2, 63^2, 67^2, 73^2, 77^2, and 83^2 fail the digit condition, so the only such four-digit perfect square is $\boxed{1369}$.

6. As no triangle can share all three sides with the nonagon, there are two remaining possibilities to consider. The first is that the triangle shares exactly 2 sides with the nonagon. These two sides of the nonagon must be adjacent, so there are 9 such triangles. The second possibility is that the triangle shares exactly 1 side with the nonagon. Once one of the 9 sides of the nonagon is selected for the triangle, the third vertex must be one of the 5 vertices that is not adjacent to the selected side, so there are $9 \cdot 5 = 45$ such triangles. Altogether, the total number of possible triangles is $9 + 45 = \boxed{54}$.

7. We find the answer by working backwards. After Emily read half the remaining pages on Saturday morning, there were $47 + 9 = 56$ pages remaining, so when she woke up on Saturday there were $2 \times 56 = 112$ pages remaining. Similarly, after she read half of the pages on Friday evening, there were $112 + 12 = 124$ pages remaining. Therefore, the total number of pages in the book was $2 \times 124 = \boxed{248}$.

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8. The ant could take a path that travels along the 40×17 face, crosses an edge of length 17, and then travels along the 17×25 face. This is the hypotenuse of a right triangle with legs of length $25+40 = 65$ and 17, and would result in a path length of $\sqrt{65^2 + 17^2} = \sqrt{4514}$. Alternatively, the ant could take a path that travels along the 40×25 face, crosses an edge of length 25, and then travels along the 25×17 face. This is the hypotenuse of a right triangle with legs of length $17 + 40 = 57$ and 25. This is slightly better, resulting in a path of length $\sqrt{57^2 + 25^2} = \sqrt{3874}$. Finally, the ant could take a path that travels along the 25×40 face, crosses an edge of length 40, and then travels along the 40×17 face. This is the hypotenuse of a right triangle with legs of length $17 + 25 = 42$ and 40. This is best, resulting in a path of length $\sqrt{42^2 + 40^2} = \sqrt{3364}$, which is exactly $\boxed{58}$.

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Team Round Solutions

1. The sum of the numbers that are even is $2+4+6+8+10+12+14+16+18+20+22+24+26+28+30 = 240$. However, the numbers that are multiples of 3, which are all of the multiples of $2 \times 3 = 6$, must be removed. The sum of those numbers is $6 + 12 + 18 + 24 + 30 = 90$. Therefore, the desired sum is $240 - 90 = \boxed{150}$.

2. The number must be 4 more than a multiple of 6 and 1 less than a multiple of 7. So there must be a multiple of 7 that is $4 + 1 = 5$ more than a multiple of 6. Since 7 is one more than 6, this first occurs at $7 \times 5 = 35$ and $6 \times 5 = 30$. Therefore, the least possible number of memes that Marlee could have is $35 - 1 = \boxed{34}$.

3. In a twelve-hour period, not including midnight, the chimes ring $1+2+3+4+5+6+7+8+9+10+11+12 = 78$ times. Since each twelve-hour period including midnight adds an additional $24-12 = 12$ rings, there are $78 * 2 + 12 = 168$ rings each day. There are 31 days in October, so the total number of rings in October is $31 \times 168 = \boxed{5208}$.

4. The figure is divided into 5 regions. Using exactly one region per triangle, there are 5 triangles. Using exactly two regions per triangle, there are 4 triangles. Using exactly three regions per triangle, there are 2 triangles. There are no triangles that use exactly four regions, and only 1 triangle that uses all five regions. Altogether, the number of triangles in the figure is $5 + 4 + 2 + 0 + 1 = \boxed{12}$.

5. The four digits sum to $2 + 4 + 7 + 9 = 22$. The greatest possible product occurs when 22 is split into two equal sums, one inside each set of parentheses. Thus, 4 must be paired with 7, and 2 with 9, for a product of $11 \times 11 = 121$. The least possible product occurs when 22 is split into the two most unequal sums. Thus 2 must be paired with 4, and 7 with 9, for a product of $6 \times 16 = 96$. The difference between the greatest and least possible products is $121 - 96 = \boxed{25}$.

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6. Suppose the original jar has a volume of 200 milliliters. Then, of that 200 milliliters, $100\% - 80\% = 20\%$, or $200 \cdot 0.20 = 40$ milliliters are vinegar. Since the second jar is half the size of the first jar, it would have a volume of $200 \div 2 = 100$ milliliters, and $100\% - 90\% = 10\%$, or $100 \cdot 0.10 = 10$ milliliters are vinegar. The resulting mixture would have a volume of $200 + 100 = 300$ milliliters, of which $40 + 10 = 50$ milliliters would be vinegar. This is $\frac{50}{300}$, which to the nearest whole number percent is $\boxed{17\%}$.

7. A four-digit palindrome can be expressed as $ABBA$, where A and B are digits. For a number to be a multiple of 99, it must be a multiple of both 9 and 11. A four-digit number $WXYZ$ is divisible by 11 when $(W + Y) - (X + Z)$ is a multiple of 11. Because $(A + B) - (B + A) = 0$, every four-digit palindrome is a multiple of 11. A number is a multiple of 9 if the sum of its digits is a multiple of 9. The sum of the digits of $ABBA$ is $2 \cdot (A + B)$. Thus, for $ABBA$ to be a multiple of 9, A and B must sum to a multiple of 9. As A cannot be 0, this only happens when AB is a two-digit multiple of 9. Therefore, the total number of four-digit palindromes that are multiples of 99 is $\boxed{10}$.

8. Consider the arrangements where Andre is seated next to Bryce. The two can be regarded as one student, so it would be as if 5 students are seated around the circular table. If the students were seated in a line there would be $5! = 120$ ways. But since the table is a circle and rotations about the table are indistinguishable, there are only $\frac{5!}{5} = 24$ possible ways. Additionally, since Andre could be on either the left or right or Bryce, there are 2 possibilities for each of these 24 ways, for a total of $2 \cdot 24 = 48$. Some of these have Charlotte on the other side of Bryce, however. Now suppose the Andre, Bryce, Charlotte trio is considered as one person. There would be $\frac{4!}{4} = 6$ ways to seat the students around the table. But the Andre, Bryce, Charlotte trio could be seated as Andre on the left of Bryce and Charlotte on the right, or Andre on the right of Bryce and Charlotte on the left, so there are $2 \cdot 6 = 12$ arrangements that have Bryce between Andre and Charlotte. Thus the number of arrangements that have Andre next to Bryce and Charlotte not next to Bryce is $48 - 12 = \boxed{36}$.

9. Let the radius of the smaller circle be r. Construct a right triangle with endpoints of the hypotenuse at centers of the circles, and one leg parallel to the bases of the trapezoid and another leg parallel to the segment of length 600. Then the shorter leg of this right triangle has a length of $200 - r$, the longer leg is has a length of $600 - 200 - r = 400 - r$, and the hypotenuse has a length of $200 + r$. By the Pythagorean Theorem, $(200 - r)^2 + (400 - r)^2 = (200 + r)^2$. This simplifies to $r^2 - 1600r + 160000 = 0$. Completing the square, $r^2 - 1600r + 640000 = 480000$, so $r - 800 = \pm 400\sqrt{3}$, and $r = 800 \pm 400\sqrt{3}$. Since r must be less than 200, $r = 800 - 400\sqrt{3}$, which to the nearest whole number is $\boxed{107}$.

10. There is 1 way for Leland to assemble a tower that is 1 unit tall. There are 2 ways for Leland to assemble a tower that is 2 units tall. Now consider a tower that is 3 units tall. If the bottom brick is 1 unit tall, the remainder of the tower can be completed in the number of ways to assemble a 2 unit tall tower, which is 2. If the bottom brick is 2 units tall, the remainder of the tower can be completed in the number of ways to assemble a 1 unit tall tower, which is 1. Therefore, the total number of ways to assemble a tower that is 3 units tall is $2 + 1 = 3$. For a tower 4 units tall, if the bottom brick is 2 units tall, there are 2 ways to assemble the remainder of the tower, and if the bottom brick is 1 unit tall, there are 3 ways to assemble the remainder of the tower, for a total of $2 + 3 = 5$. The pattern continues, so that a tower 5 units tall can be assembled in $5 + 3 = 8$ ways, a tower 6 units tall can be assembled in $8 + 5 = 13$ ways, and so on. These are Fibonacci numbers, and the sequence in this instance is $1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987$, so the number of ways to assemble a tower that is fifteen units tall is $\boxed{987}$.

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